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# 13 HAZARD TO LIFE

## 13.1 Introduction

13.1.1 This section evaluates the hazard to life impacts arising from any existing hazardous facilities with their consultation zones encroaching upon the Project Site, and any planned hazardous facilities proposed by the Project. The requirements for assessing hazard to life are outlined in Annex 4 of the Technical Memorandum on Environmental Impact Assessment Process (EIAO-TM) and Section 3.4.13 and Appendix L of the EIA Study Brief (ESB-363/2023).

## 13.2 Environmental Legislation, Standards and Guidelines

13.2.1 The estimated risk levels shall be compared with the individual and societal risk criteria set out in the Hong Kong Risk Guidelines (HKRG) to determine the acceptability of the risk levels.

*Hong Kong Risk Guidelines*

13.2.2 Individual risk is the predicted increase in the chance of fatality per year to an individual due to a potential hazard. The individual risk guidelines require that the maximum level of off-site individual risk should not exceed 1 in 100,000 per year i.e. $1\times10^{-5}$ per year.

13.2.3 Societal risk refers to the risks to the whole population. It is expressed in terms of lines plotting the cumulative frequency (F) of number of fatalities (N) or more deaths in the off-site population from incidents at the installation against N. Two F-N risk lines are used in the HKRG that demark “Acceptable” or “Unacceptable” societal risks. To avoid major disasters, there is a vertical cut-off line at the 1,000 fatality level extending down to a frequency of 1 in a billion years (i.e. $1\times10^{-9}$ per year). The intermediate region indicates that the acceptability of societal risk is borderline and should be reduced to a level which is “as low as reasonably practicable” (ALARP). It seeks to ensure that all practicable and cost-effective measures that can reduce risk are considered. The HKRG is presented graphically in **Exhibit 13.1**.

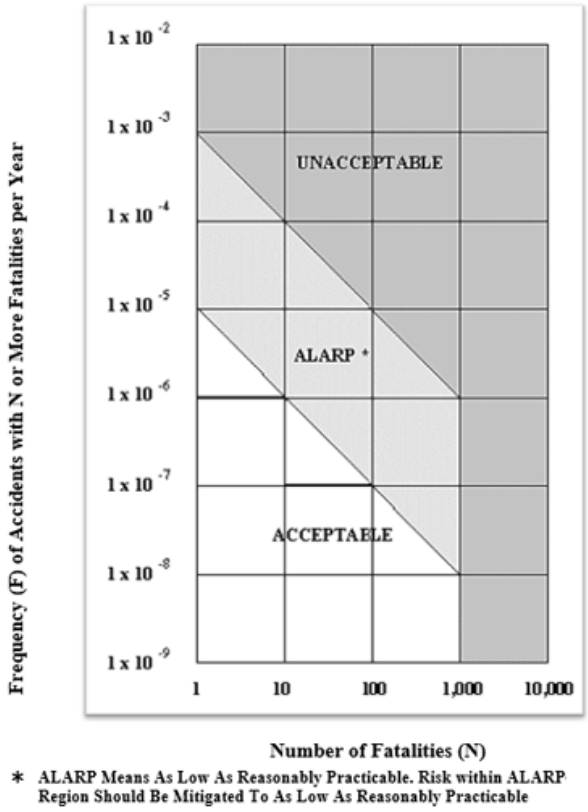



**Exhibit 13-1 Societal Risk Guidelines**

## 13.3 Evaluation of Potential Hazard to Life Impacts

13.3.1 According to the EIA Study Brief, hazard to life assessment would be required to evaluate the potential impacts due to the existing and/or planned hazardous facilities (if any) during construction and operational stages of the Project.

13.3.2 Ngau Tam Mei Water Treatment Works (NTMWTW) is situated at approximately 170 m east from the Development Area. As advised by Water Supplies Department, there is no liquid chlorine storage in NTMWTW. Therefore, NTMWTW is not considered as hazardous facility, and there would not be any hazards related to storage of liquid chlorine.

13.3.3 According to the Recommended Outline Development Plan of the Project, there are no planned hazardous facilities within the Development Area. Therefore, no hazard to life impact would be arisen from the operation of the Project.

## 13.4 Mitigation Measures

13.4.1 Mitigation measures are deemed not necessary as no hazard to life impact is anticipated.

## 13.5 Conclusion

13.5.1 No hazard to life impact is anticipated as there are no existing hazardous facilities with their consultation zones encroaching upon the Project Site and no planned hazardous facilities proposed under the Project.